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Physiology-guided management of patients with severe hypoxemia ineligible for ECMO: a multidisciplinary lung rescue team approach

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Abstract

Background ARDS affects a significant proportion of ventilated ICU patients and carries high mortality, especially in cases of severe hypoxemia unresponsive to standard treatments and ineligible for extracorporeal membrane oxygenation (ECMO). In such situations, care often shifts to comfort measures due to limited data and structured guidance. To address this gap, Massachusetts General Hospital established the Lung Rescue Team (LRT), a multidisciplinary group using advanced physiological tools to provide individualized, real-time management beyond protocolized care.

Objective To evaluate the feasibility and application of a complex, physiology-guided framework in patients with severe hypoxemia who had not responded to standard or advanced interventions and were ineligible for rescue therapies such as ECMO, population typically excluded from clinical trials and often regarded as beyond curative treatment.

Design This single-center, retrospective observational study included adult patients in the intensive care unit (ICU) ($n = 58$) with severe hypoxemia not eligible for ECMO who were evaluated by the LRT at Massachusetts General Hospital between 2014 and 2025. Multimodal physiological assessments were performed which included electrical impedance tomography (EIT), esophageal manometry (EM), and transthoracic echocardiography (TTE). Interventions were categorized and clinical outcomes assessed.

Results LRT consultation led to changes in respiratory management in 82.3% of patients who had already attend to receive maximal therapy, most commonly adjustments to PEEP based on EIT or transpulmonary pressure. The median ICU length of stay was 18.5 days (IQR 10.5–31.5). Survival was 48.3% at 90 days and 44.8% at 1 year. The leading cause of death was multiorgan failure (50%), followed by hypoxemia (25%) and unresponsive shock (25%).

Conclusions This study describes the experience of a physiology-guided, multidisciplinary LRT in patients with severe hypoxemia ineligible for ECMO, individuals often excluded from clinical trials and frequently considered beyond curative treatment. Survival in this cohort was far from negligible, highlighting the value of systematic re-evaluation rather than premature transition to palliation. Although causal inferences cannot be drawn from this

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retrospective, single-center analysis, the findings provide insight into an underreported population and generate hypotheses for future prospective studies.

Keywords Respiratory insufficiency, Severe hypoxemia, Standard of care, Lung rescue team, Individualized treatment, Electrical impedance tomography, Esophageal manometry, ECMO

Introduction

Acute respiratory distress syndrome (ARDS) affects approximately 20% of mechanically ventilated patients and is associated with mortality rates of up to 40% in its most severe forms, even when standard therapies are implemented [1–4]. Severe hypoxemia defined as a sustained $\text{PaO}_2/\text{FiO}_2$ ratio < 100 [5–7], represents a devastating condition in the intensive care unit (ICU) [1–4]. When hypoxemia persists and patients are not eligible for extracorporeal membrane oxygenation (ECMO) this mortality may rise as high as 86% [8].

Therapies for ARDS include mechanical ventilation, deep sedation, neuromuscular blockade, and more advanced interventions such as recruitment maneuvers, prone positioning, and inhaled pulmonary vasodilators. When all these options fail or cannot be implemented, the final recourse is the use of rescue therapies such as ECMO [9, 10].

When advanced therapies are unsuccessful and ECMO is not an option, care frequently shifts from curative to comfort-focused approaches [11]. However, there is limited longitudinal data describing the outcomes of patients who continue to experience severe hypoxemia despite the application of standard and advanced therapeutic interventions [11, 12]. In such cases, clinical management is often guided by assumptions of futility rather than grounded in physiological understanding, leaving clinicians without structured frameworks to guide ongoing care.

In the United States, mechanical ventilation is primarily managed by respiratory therapists in conjunction with intensivists following institutional protocols informed by international guidelines. The different strategies have change along the years and emphasize lung-protective strategies, including low tidal volumes (6–8 mL/Kg predicted body weight), plateau pressures < 30 cmH₂O, driving pressures < 15 cmH₂O, and PEEP titration guided by the ARDSNet PEEP/ FiO_2 Tables [8, 13, 17]. While evidence-based, these protocols may fall short in patient that present physiologically complex cardiopulmonary interactions or non-standard respiratory mechanics.

In response to the need for individualized ventilatory management, the Massachusetts General Hospital (MGH) established the Lung Rescue Team (LRT) in 2014. This multidisciplinary team, consisting of a critical care physician, ICU fellow, respiratory therapist, and ICU nurse, was designed to support decision-making in patients with severe hypoxemia. From its inception, the

LRT integrated multimodal cardiopulmonary assessment tools, driven by the team's clinical expertise and the complexity of the patient population, including electrical impedance tomography (EIT), esophageal manometry (EM), and transthoracic echocardiography (TTE) to evaluate ventilation, perfusion and heart-lung interaction in real time [18–23]. The combination of available physiologic tools and experience enabled dynamic, patient-specific evaluations beyond protocolized care.

We designed a longitudinal descriptive study of patients with severe hypoxemia who had failed standard and advanced interventions and were not eligible for rescue therapies such as ECMO, a population typically excluded from clinical studies and often considered beyond curative intervention. Our aim was to evaluate whether an advanced physiology-guided, multidisciplinary LRT could provide actionable data to support individualized ventilatory and hemodynamic adjustments, challenge assumptions of futility, and guide meaningful decision-making by the care team and families. The primary outcomes were the implementation rate of LRT recommendations and changes in ventilator management, while secondary outcomes included time from intubation to LRT consultation, ICU length of stay, duration of mechanical ventilation, and survival at ICU discharge.

Methods

Study design and setting. This was a single-center, retrospective observational study conducted at the MGH, a tertiary academic medical center affiliated with Harvard Medical School in Boston, Massachusetts. The study was approved by the Mass General Brigham Institutional Review Board (Protocol number #2020P003196).

Inclusion and exclusion criteria. Adult patients (≥ 18 years) with severe hypoxemia, $\text{PaO}_2/\text{FiO}_2$ ratio ≤ 100 , who were evaluated at Massachusetts General Hospital (MGH) deemed ineligible for ECMO by primary care or ECMO team, and who were referred to the LRT between January 2014 and February 2025, were included in the study. Patients who died after the referral initiation and prior to LRT assessment were excluded from the analysis.

Lung rescue team and workflow. The Lung Rescue Team (LRT) is a multidisciplinary group established in 2014. Activation of the LRT can occur through two distinct pathways, both initiated by the ICU team, respiratory therapist, or nurse practitioner. The first involves patients with complex clinical conditions, such as obesity,

interstitial lung disease, or newly emerging diseases (e.g., COVID-19), or when a general call for specialized assistance is made. The second occurs when patients remain severely hypoxemic despite receiving both conventional and rescue therapies. In these cases, the LRT conducts structured bedside evaluations and provides treatment recommendations based on individualized physiological assessments.

Data collection and variables. Data were extracted from the electronic medical record and the LRT database. Collected variables included demographics, comorbidities, reason for intubation, illness severity scores (APACHE II and SOFA) and physiologic data (pH, PaCO₂, FiO₂, PO₂ and PF ratio), ventilator settings (specially ventilatory ratio) and respiratory mechanics before and after LRT evaluation, and all LRT interventions.

Assessment and interventions. The LRT utilize one or more of the following tools for physiological evaluation: (I) EIT (Timpel® ENLIGHT 2100), used for recruitment maneuvers, PEEP titration, ventilation-perfusion matching, and positioning strategies. (II) EM, used for assessment of transpulmonary pressures and PEEP adjustments. (III) TTE, used for cardiac function assessment and cardiopulmonary interaction. Over the years, these tools have undergone continuous development, including enhancements in software and equipment. Their use during each assessment was guided by the expertise and clinical judgment of the LRT member conducting the evaluation. The LRT's clinical recommendations were categorized into six groups: (I) No change from the current management, (II) Adjustment of ventilator settings (PEEP, tidal volume, respiratory rate), (III) Alterations in body position (e.g., transitioning from supine to prone or vice versa) and adjustments in bed angle (0°, 10°, or 30°), (IV) Initiation or adjustment of iNO, (V) Recommendations for bronchoscopy or TTE (further cardiological evaluation), (VI) Initiation or discontinuation of neuromuscular blockade (Fig. 1).

Outcomes. Primary outcomes included the implementation rate of LRT recommendations and changes in ventilator management. Secondary outcomes included time from intubation to LRT consultation, ICU length of stay, duration of mechanical ventilation, and survival at ICU discharge, 28 days, 90 days, and one year.

Statistical analysis. Descriptive statistics were used to summarize the data. Continuous variables are presented as medians with interquartile ranges (IQRs), and categorical variables are expressed as counts and percentages. Survival was presented as a Kaplan Meier curve. Statistical analyses were performed using GraphPad Prism (GraphPad Software, Inc.10.4.1 (627)).

Results

Study population. Among 396 patients evaluated by the LRT between January 2014 and February 2025, 58 met the inclusion criteria for refractory hypoxemia and were not eligible for ECMO. Patients who died prior to LRT evaluation were excluded (Fig. 2).

Patient characteristics. The median age was 52 years (IQR 36–65.3), and 32.8% were female. The median body mass index (BMI) was 35.5 Kg/m² (IQR 28.2–47.8), with 67.2% of patients classified as obese (BMI ≥ 30 kg/m²). The most common comorbidities were hypertension (58.6%), diabetes (43.1%), and a history of smoking (43.1%).

Reasons for intubation included COVID-19-related ARDS (27.6%) and non-COVID-19-related ARDS (29.3%), followed by septic shock without ARDS (15.5%), cardiovascular failure (8.6%), neurological compromise (6.9%), trauma (3.5%), and other causes (8.6%) (Table 1).

Severity of illness and pre-LRT interventions. The median time from intubation to LRT consultation was 3 days (IQR 1–10.5). Median APACHE II and SOFA scores were 26 (IQR 19–32) and 13 (IQR 11–16), respectively. At the time of LRT assessment, the median PaO₂/FiO₂ ratio was 89 (IQR 76–96), FiO₂ was 1.0 (IQR 0.8–1.8), PaCO₂ was 51 mmHg (IQR 45.7–63), pH was 7.32 (IQR 7.25–7.38), and the ventilatory ratio was 2.19 (IQR 1.79–2.53), all of which reflect a high severity of illness (Table 2).

All patients underwent recruitment maneuvers and decremental PEEP trials following institutional protocols before LRT assessment (*supplementary material "Recruitment maneuver & MGH Decremental PEEP Trial"*). Neuromuscular blockade was used in 91.4% of patients; in the remaining 8.6%, the ICU team judged the patient to be adequately synchronous with the ventilator. Prone positioning had been applied in 41.4%, while 58.6% had not been prone due to relative or absolute contraindications. Absolute contraindications included unstable spinal fracture; relative contraindications included hemodynamic instability, unstable pelvic or long bone fracture, open abdomen, obesity, and illness that lead on high abdominal hypertension [22]. Inhaled nitric oxide (iNO) was administered in 39.7% of patients at the time of consultation. Although all patients had been previously deemed ineligible for ECMO (*supplementary material "MGH ECMO criteria"*), following LRT assessment, 4 patients (6.9%) were ultimately cannulated (Table 3).

Ventilatory mechanics and LRT recommendations. Driving pressure was 14 cmH₂O (IQR 11–17), compliance was 26 mL/cmH₂O (IQR 20–37), and PEEP set by the ICU team was 14 cmH₂O (IQR 10–15) (*supplementary material "Table 1"*).

The LRT performed a single assessment per patient, resulting in a total of 58 assessments, each corresponding to the initial evaluation within this population. Of these,

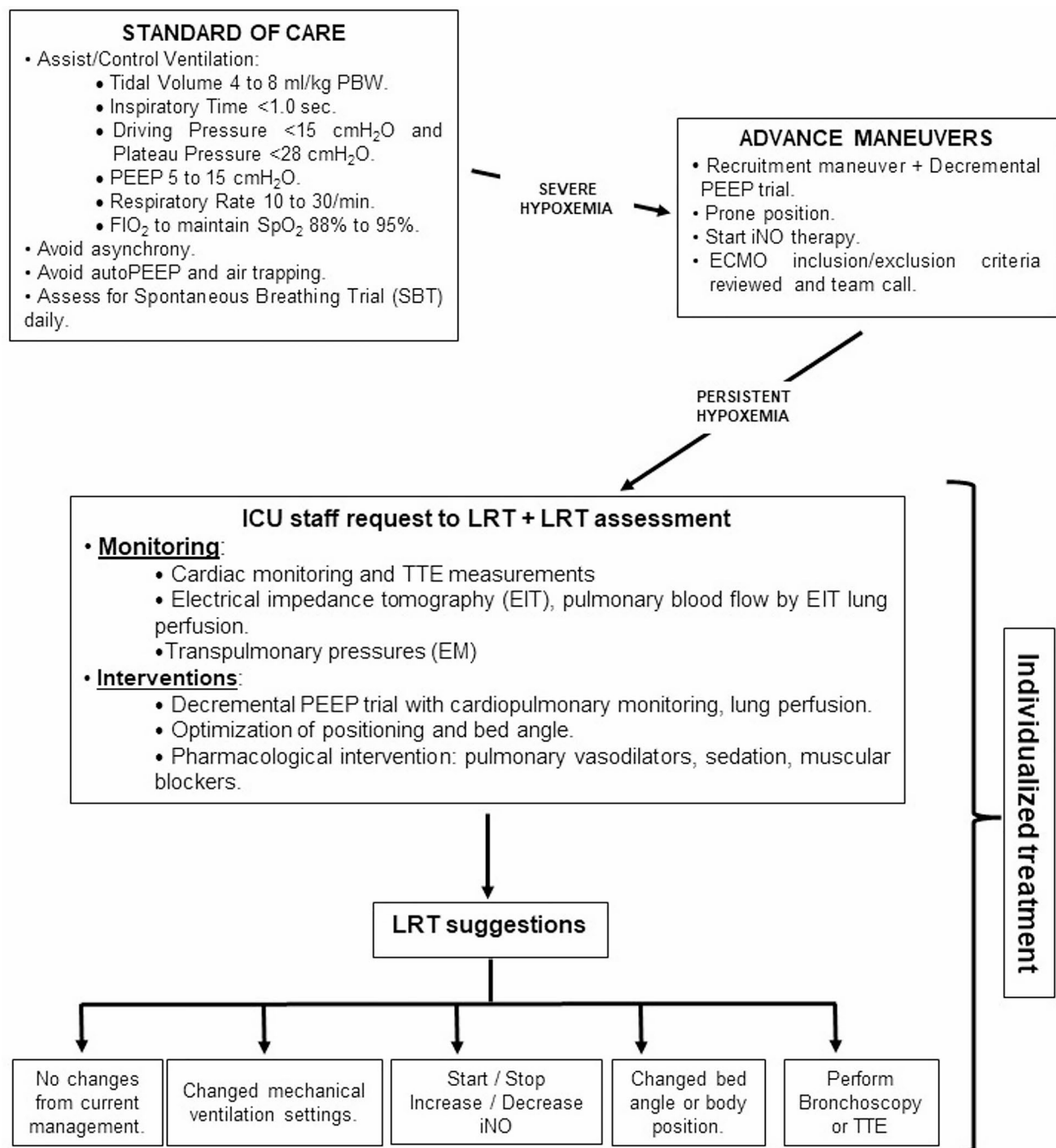


Fig. 1 Flowchart of enrolment assessments and suggestions from LRT. PBW predicted body weight, PEEP positive end expiratory pressure, FIO₂ fraction of inspired oxygen, SPO₂ peripheral oxygen saturation, iNO inhaled nitric oxide, RV right ventricle, ICU intensive care unit, LRT lung rescue team, IRB institutional review board, TTE transthoracic echocardiography

EIT was used in 44 assessments (55.7%), supporting recruitment maneuvers ($n=19$), PEEP titration ($n=39$), ventilation/perfusion analysis ($n=27$), and patient positioning ($n=6$). Esophageal manometry was employed for PEEP titration in 8 cases and for transpulmonary pressure measurement in 36 (Fig. 2). The choice between one tool or another was based on the expertise and specific

roles of the LRT members. Notably, none of the LRT interventions were discontinued due to complications or hemodynamic instability at the time of assessment.

From the total assessments ($n=58$), 79 recommendations were made by the LRT and in 82.3% ($n=65$) of these recommendations, a change in management was suggested, most commonly involving adjustments to

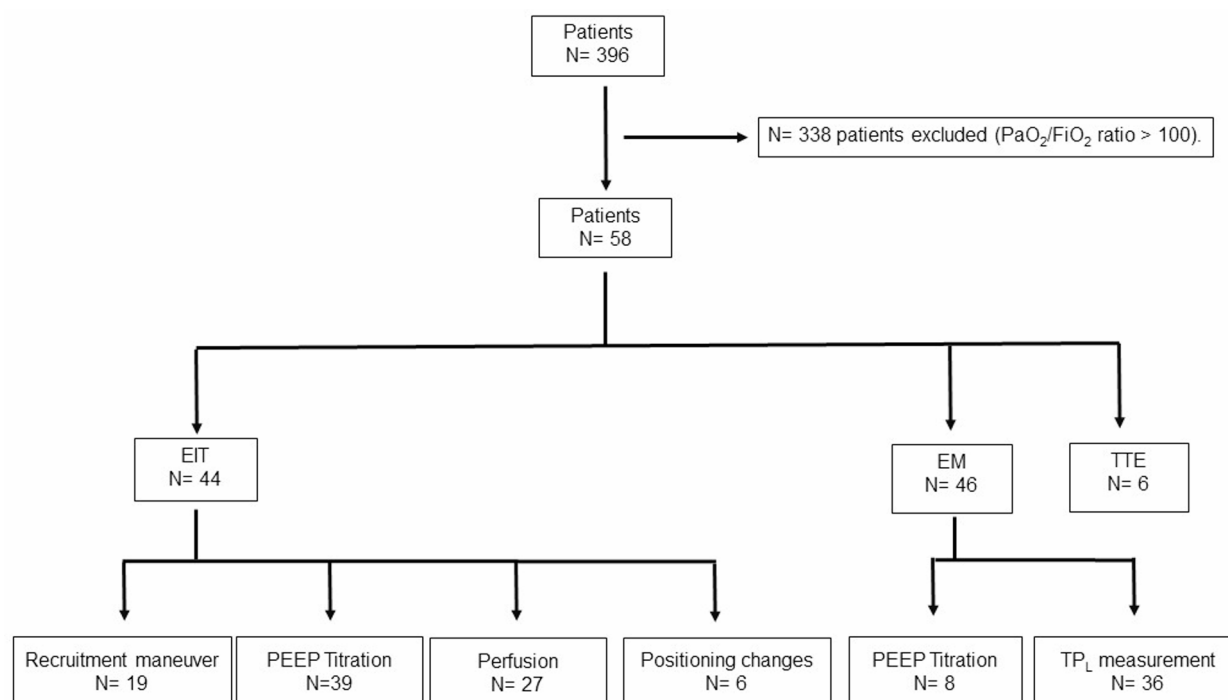


Fig. 2 Study population and LRT assessment. PaO₂ partial pressure of oxygen, FiO₂ fraction of inspired oxygen, EIT electrical impedance tomography, EM esophageal manometry, TTE transthoracic echocardiography, PEEP positive end expiratory pressure, TP_L transpulmonary pressure

mechanical ventilation settings (62.0%), particularly modifications to PEEP (87.8% of those 49 adjustments to mechanical ventilation settings). Other recommendations included body position adjustments (8.9%), initiation or up-titration of iNO (6.3%), and performance of bronchoscopy or TTE (3.8%) (Table 4). Clinical teams maintained after implemented LRT recommendations for at least 24 h in 82.8% of cases (supplementary material “Table 1”).

Clinical outcomes. Among all patients, 27 (46.6%) survived to ICU discharge, 37 (63.8%) were alive at 28 days, 28 (48.3%) at 90 days, and 26 (44.8%) at 1 year (Table 5 and Fig. 3). The leading cause of death was multiorgan failure (50%), followed by hypoxemia (25%) and unresponsive shock (25%). The median duration of mechanical ventilation was 13.5 days (IQR 6.7–27), and the median ICU length of stay was 18.5 days (IQR 10.5–31.5) (Table 5).

Discussion

This study showed that among 58 patients with severe hypoxemia who were not eligible for ECMO, 27 (46.6%) survived to ICU discharge, 37 (63.8%) were alive at 28 days, 28 (48.3%) at 90 days, and 26 (44.8%) at 1 year. In clinical practice, this group is often considered beyond rescue, and care frequently transitions toward palliation. Our findings suggest that structured bedside re-evaluation by a physiology-guided, multidisciplinary LRT can lead to actionable changes in ventilatory and

hemodynamic management and provide meaningful opportunities for recovery, even in patients typically regarded as “non-salvageable.” [10]. The survival rates observed in this cohort are noteworthy given the severity of hypoxemia and the exclusion from extracorporeal rescue. Although these outcomes should not be interpreted as causal effects of LRT interventions, they highlight that systematic, physiology-guided re-assessment may counter therapeutic nihilism and help identify treatment strategies with tangible benefit. Importantly, LRT consultations facilitated renewed clinical engagement for both care teams and families, helping to reframe decisions that otherwise might have been directed toward withdrawal of support.

The LRT approach centered on multimodal physiological assessment. From its inception in 2014, the LRT incorporated EIT, EM, and TTE as core bedside tools. EIT was the most frequently employed, used in more than half of all evaluations, providing real-time assessment of regional ventilation patterns that guided PEEP and tidal volume adjustments. This technology has been available for over 40 years, and its use and clinical understanding have significantly increased over the past two decades [24, 25]. EM enabled estimates of transpulmonary pressures to individualize ventilatory settings, while TTE allowed evaluation of right and left heart function and heart-lung interactions. Together, these complementary modalities enabled individualized ventilatory and

Table 1 Demographics

	Lung rescue team cohort
Patients, n	58
Sex, women, n (%)	19 (32.8)
Age, years, median (IQR)	52 (36–65.3)
Race	
White, n (%)	32 (55.2)
Black or African American, n (%)	11 (19.0)
Asian, n (%)	1 (1.7)
American Indian or Alaska native, n (%)	1 (1.7)
Other, n (%)	2 (3.5)
Declined, n (%)	1 (1.7)
Unavailable, n (%)	10 (17.2)
Ethnic group	
Hispanic or Latino, n (%)	11 (19.0)
Not Hispanic or Latino, n (%)	40 (68.9)
Unavailable, n (%)	7 (12.1)
BMI, Kg/m², median (IQR)	
	35.5 (28.2–47.8)
Normal weight, BMI 18.5–24.9, n (%)	4 (6.9)
Overweight, BMI 25–29.9, n (%)	15 (25.9)
Obesity class 1, BMI 30–34.9, n (%)	10 (17.2)
Obesity class 2, BMI 35–39.9, n (%)	6 (10.3)
Obesity class 3, BMI ≥ 40, n (%)	23 (39.7)
Comorbidities	
None, n (%)	1 (1.7)
Hypertension, n (%)	34 (58.6)
Hyperlipidemia, n (%)	17 (29.3)
Diabetes, n (%)	25 (43.1)
Hypothyroidism, n (%)	7 (12.1)
Myocardial infarction/coronary heart disease, n (%)	10 (17.2)
CHF, n (%)	12 (20.7)
AF, n (%)	9 (15.5)
Cerebrovascular disease, n (%)	3 (5.2)
Dementia, n (%)	0 (0)
Smoking (actual or former), n (%)	25 (43.1)
COPD, n (%)	6 (10.3)
Asthma, n (%)	7 (12.1)
OSA, n (%)	12 (20.7)
CKD, n (%)	11 (19.0)
Liver disease, n (%)	10 (17.2)
Malignancy, n (%)	9 (15.5)
Immune deficiency, n (%)	6 (10.3)
Connective tissue disease, n (%)	0 (0)
PVD, n (%)	4 (6.9)
Other, n (%)	43 (74.1)
Reason for intubation	
ARDS COVID related, n (%)	16 (27.6)
ARDS no COVID related, n (%)	17 (29.3)
Septic shock no pulmonary related, n (%)	9 (15.5)
Cardiovascular failure, n (%)	5 (8.6)
Neurological status, n (%)	4 (6.9)

Table 1 (continued)

	Lung rescue team cohort
Trauma-related cause, n (%)	2 (3.5)
Other causes, n (%)	5 (8.6)

IQR interquartile range, BMI body mass index, CHF congestive heart failure, AF atrial fibrillation, COPD chronic obstructive pulmonary disease, OSA obstructive sleep apnea, CKD chronic kidney disease, PVD peripheral vascular disease, ARDS acute respiratory distress syndrome

circulatory strategies, all delivered without major complications or hemodynamic instability.

Previous studies have largely focused on patients eligible for ECMO or on cohorts with less severe hypoxemia. Few data exist for patients who fail conventional and rescue interventions but are excluded from extracorporeal support. In these scenarios, guidelines offer limited direction, and practice is heterogeneous. Our findings contribute to this knowledge gap by describing outcomes and management in a real-world population frequently overlooked in trials. Importantly, as emphasized by others, retrospective data cannot establish causality but can illuminate outcomes and physiological characteristics of patients who fit criteria for neither conventional (i.e., lung-protective, low-stretch ventilation) nor advanced management. This underscores a broader lesson: stratification of ARDS patients based solely on oxygenation may fail to predict clinical trajectory and risks misleading decision-making. Our results align with this perspective, illustrating that systematic, physiology-based reassessment can uncover therapeutic opportunities in patients otherwise considered beyond rescue. For the same reasons, conventional severity scoring systems, such as the Sequential Organ Failure Assessment (SOFA) and the Acute Physiology and Chronic Health Evaluation II (APACHE II), may have limited utility in this context. These scores capture initial organ dysfunction but are not designed to inform ventilator management or evaluate the potential for physiologic responsiveness in real time. Our approach represents a shift from static scoring toward dynamic bedside physiology [15, 18, 19, 22, 26].

The retrospective nature of this study, spanning more than a decade, necessarily reflects evolving standards of ARDS management. Prone positioning, recruitment strategies, and the broader adoption of EIT and EM have developed over this timeframe. Although the frequency and sophistication of tool use increased with growing institutional expertise, EIT, EM, and TTE were available and applied since the implementation of the LRT in 2014. Although this evolution may have influenced management and outcomes, our objective was not to compare results across different eras but to describe the overall feasibility and impact of a multidisciplinary LRT. In fact, the consistency of the LRT approach across diverse practice conditions reinforces the potential robustness

Table 2 Key clinical characteristics at initial evaluation

	n = 58
Length between intubation and assessment, days, median (IQR)	3 (1–10.5)
Sever of illness	
APACHE II at the time of ICU admission, n, median (IQR)	26 (19–32)
SOFA at assessment, n, median (IQR)	13 (11–16)
PaO ₂ /FiO ₂ ratio, n, median (IQR)	89 (76–96)
PaCO ₂ , mmHg, n, median (IQR)	51 (45.7–63)
pH, n, median (IQR)	7.32 (7.25–7.38)
FiO ₂ , n, median (IQR)	1.0 (0.8–1.0.8.0)
Ventilatory ratio, n, median (IQR)	2.19 (1.79–2.53)
Length of sever hypoxemia before the assessment, hours, median (IQR), [min, max]	9.5 (4.7–16.5) [1–53]
APACHE acute physiologic assessment and chronic health evaluation scoring, IQR interquartile range, SOFA sepsis-related organ failure assessment	

Table 3 Key clinical characteristics at initial evaluation

	n = 58
Use of rescue interventions	
Recruitment maneuver + Decremental PEEP trial , n (%)	58 (100)
Prone position before the assessment, n (%)	24 (41.4)
Not prone for absolute contraindications (*), n (%)	1 (1.7)
Not prone for relative contraindications (**), n (%)	33 (56.9)
Patients under iNO at the time of the assessment, n (%)	23 (39.7)
iNO not used because non-responders or ICU team did not feel useful, n (%)	35 (60.3%)
Use of neuromuscular blockers , n (%)	53 (91.4)
No neuromuscular blockers were used as the team felt the patient was already synchronous with the vent and did not require further use, n (%)	5 (8.6)
ECMO declined before assessments, n (%)	58 (100)
ECMO declined by the primary ICU team, n (%)	18 (33.3)
ECMO declined by the ECMO team, n (%)	36 (66.7)
ECMO started after assessments, n (%)	4 (6.9)

* Absolute contraindication: unstable spinal fracture

**Relative contraindications: hemodynamic instability, unstable pelvic or long bone fracture, open abdomen, obesity and illness leading abdominal hypertension

PEEP positive end expiratory pressure, iNO inhaled nitric oxide, ECMO extracorporeal membrane oxygenation

of this model. Several studies conducted by our group during this period further refined our understanding of ventilation-perfusion mismatch, patient selection, safety metrics in physiology-guided [22, 27–29]. This study highlights the limitations of standardized protocols in complex pathophysiological states and supports the utility of personalized approaches driven by real-time physiological data [30].

Several limitations must be acknowledged. First, this was a small and retrospective analysis from a single tertiary care center cohort, and as such, our findings may

Table 4 LRT suggestion

	n = 79
Changes in mechanical ventilation settings, n (%)	49 (62.0)
Changes in PEEP level, n (%)	43 (87.8) n = 49
Changes in tidal volume, n (%)	5 (10.2) n = 49
Change in respiratory rate, n (%)	1 (2.0) n = 49
Changes in body position or bed angle, n (%)	7 (8.9)
Perform Bronchoscopy or TTE, n (%)	3 (3.8)
Start or increase iNO, n (%)	5 (6.3)
Stop neuromuscular blockers, n (%)	1 (1.3)
No changes from the current management, n (%)	14 (17.7)
LRT lung rescue team, PEEP positive end expiratory pressure, TTE transthoracic echocardiography, iNO inhaled nitric oxide	

Table 5 Outcomes

Outcomes	n = 58
Tracheostomy, n (%)	16 (27.6)
Length of intubation, days, median (IQR)	13.5 (6.7–27)
Length of ICU stay, days, median (IQR)	18.5 (10.5–31.5)
Alive during ICU stay, n (%)	27 (46.6)
Alive at 28 days, n (%)	37 (63.8)
Alive at 90 days, n (%)	28 (48.3)
Alive at 365 days, n (%)	26 (44.8)
Cause of death n = 32	
Hypoxemia, n (%)	8 (25)
Unresponsive shock, n (%)	8 (25)
Multiorgan failure, n (%)	16 (50)
ICU intensive care unit, IQR interquartile range	

not be generalizable to other settings with different ICU structures or less familiarity with advanced ventilatory tools. Second, the extended duration of the study period (about 11 years) introduces potential heterogeneity, as standards in the management of ARDS have evolved significantly over time. Changes in practices such as prone positioning, the use of EIT, EM, TTE, and referral patterns for advanced therapies may have influenced clinical decision-making. Nonetheless, it is important to emphasize that the primary objective of this study was not to compare outcomes across eras but to assess the feasibility and clinical impact of implementing an LRT for patients with severe hypoxemia who were not candidates for ECMO. Third, as said before in this piece of work, the lack of a control group limits causal inference regarding the impact of LRT interventions on patient outcomes. Additionally, it should also be acknowledged as a limitation that not all patients who were denied ECMO were captured, only those for whom the LRT was consulted, which may have led to an underestimation of mortality rates. Finally, decisions to implement LRT recommendations were made by the primary ICU teams, potentially introducing variability based on individual clinical judgment and institutional culture. Despite these limitations, this study challenges the assumption that nothing more can be done for patients with severe hypoxemia after all

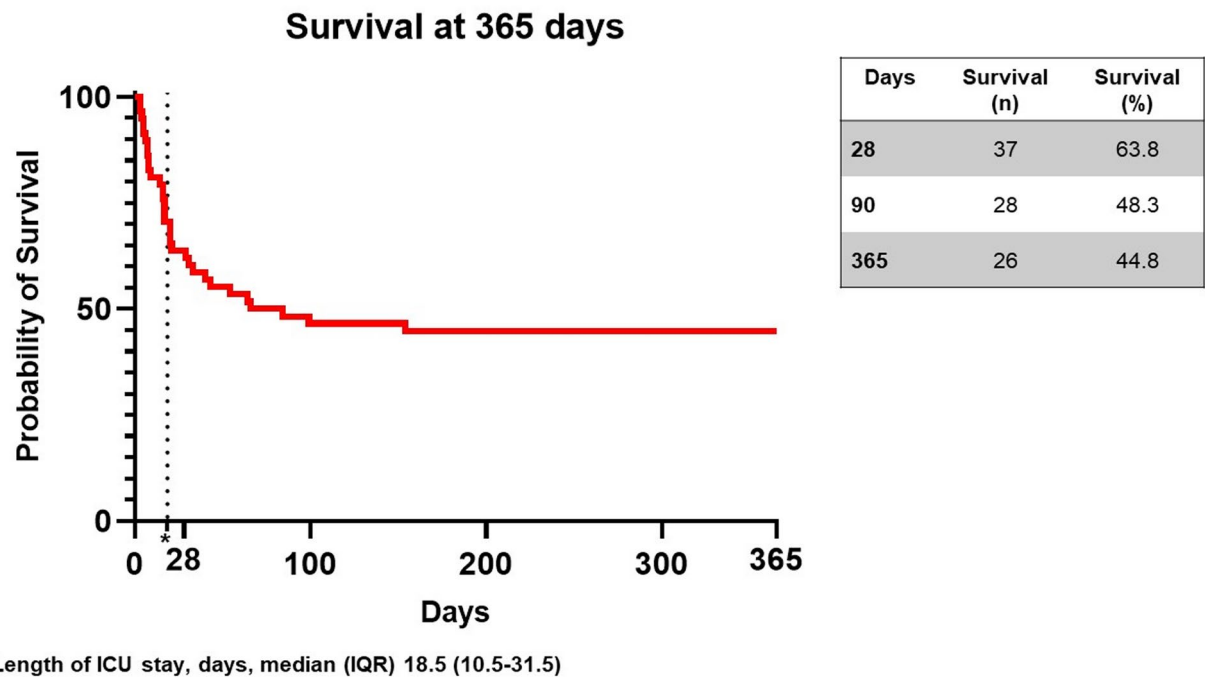


Fig. 3 Kaplan Meier 365 days survival of LRT cohort

available therapies have been attempted. The LRT model may serve as a scalable strategy to facilitate the use of advanced physiologic-guided tools and promote individualized ventilation strategies in patients who are unresponsive to conventional therapies.

In conclusion, this study presents our group’s experience in managing patients with acute respiratory failure and severe hypoxemia, individuals often excluded from interventional trials and rarely monitored longitudinally once conventional therapies fail. Based on our observations, the use of advanced, physiological, and individualized monitoring tools such as EIT, EM, and TTE appears to offer a structured approach for meaningful clinical intervention in these complex cases. The implementation of the LRT appears to enable a higher degree of personalized care in severe hypoxemia. This study should be viewed as hypothesis-generating, prompting future investigation into whether such an approach can improve morbidity and mortality in this high-risk population. Prospective multicenter studies are needed to test these hypotheses and to assess the broader impact of this strategy in critical care practice.

Supplementary Information
The online version contains supplementary material available at <https://doi.org/10.1186/s13054-025-05709-9>.

Supplementary Material 1.

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Author contributions
FRS: Conceived and designed the study; acquired, analyzed and interpreted the data; drafted the initial work.CM: Acquired and analyzed the data; reviewed critically the manuscript.GA: Acquired and analyzed the data; reviewed critically the manuscript.RRDSS: Acquired, analyzed, and interpreted the data; reviewed critically the manuscript.SS: Acquired the data; reviewed critically the manuscript.BSF: Acquired the data; reviewed critically the manuscript.TG: Acquired the data; reviewed critically the manuscript.EAB: Reviewed critically the manuscript.CJLV: Reviewed critically the manuscript.MC: Reviewed critically the manuscript.LB: Conceived and designed the study; acquired, analyzed and interpreted the data; reviewed critically the manuscript.

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Data availability
Upon request to the corresponding author.

Declarations

Ethics approval and consent to participate
MGB IRB Protocol #: 2019P001995. No patient consent was obtained as this was a prospective, non-interventional, observational study.

Consent for publication
All authors have reviewed and approved the final version of the manuscript for publication and collectively accept responsibility for the integrity and accuracy of the work.

Competing interests

The authors declare no competing interests.

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